

Drive for better vision



OTA BootFromI2C User Guide V1.2

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Agenda

- **Check Before Starting BootFromI2C**
- **BootFromI2C Operations**
- **OTA Upgrade Operations**

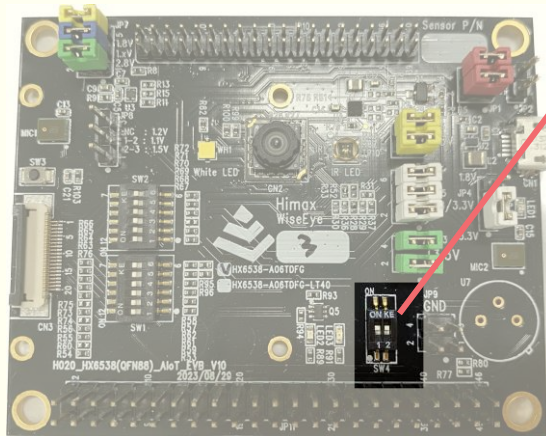
Check Before Starting BootFromI2C

Check Before Starting BootFromI2C

1. Do the EVB and already set to the Flash mode?

❖ If not, two steps to set Flash mode:

- ① Turn off the EVB power.
- ② Set the switches on EVB: **SWITCH [1, 2] = [ON, OFF]**



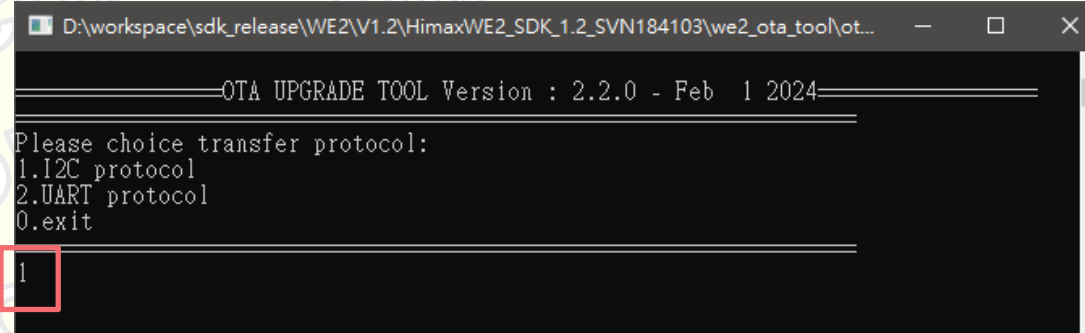
SWITCH SETTING	MODE
 1 2 [ON, OFF]	EVB: Flash mode

BootFromI2C Operations

BootFromI2C Operations (1/4)

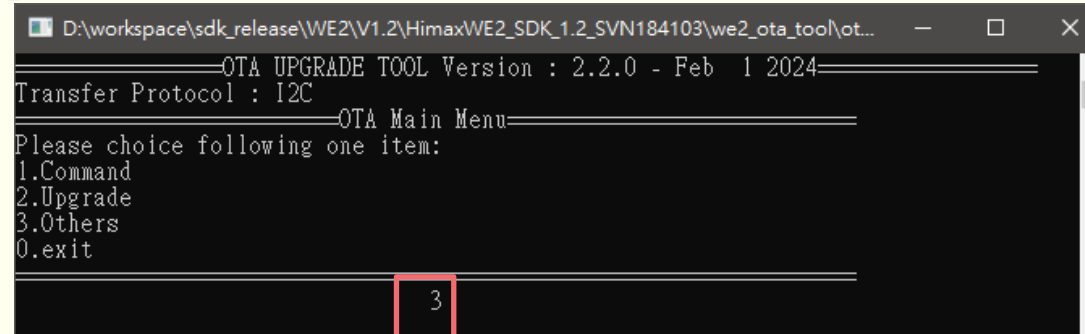
- Turn on the power of EVB
- Launch OTA tool
(path: SDK_root/we2_ota_tool/ota.exe)
- Operate by following steps:

① Key in “1” (1. I2c protocol)



```
D:\workspace\sdk_release\WE2\V1.2\HimaxWE2_SDK_1.2_SVN184103\we2_ota_tool\ot...  
=====OTA UPGRADE TOOL Version : 2.2.0 - Feb  1 2024=====  
Please choice transfer protocol:  
1.I2C protocol  
2.UART protocol  
0.exit  
1
```

② Key in “3” (3. Others)



```
D:\workspace\sdk_release\WE2\V1.2\HimaxWE2_SDK_1.2_SVN184103\we2_ota_tool\ot...  
=====OTA UPGRADE TOOL Version : 2.2.0 - Feb  1 2024=====  
Transfer Protocol : I2C  
=====OTA Main Menu=====  
Please choice following one item:  
1.Command  
2.Upgrade  
3.Others  
0.exit  
3
```

BootFromI2C Operations (2/4)

- ③ Key in “1” (1. Boot From I2C (Application reboot to 2nd loader))

```
D:\workspace\sdk_release\WE2\V1.2\HimaxWE2_SDK_1.2_SVN184103\we2_ota_tool\ot...
=====OTA UPGRADE TOOL Version : 2.2.0=====
=====Other Tools Menu=====
Please choice following one item:
1.Boot from I2C (Application reboot to 2nd loader).....[BOOTROM]
2.Downloader FW via I2CBoot (Application).....[BOOTLOADER]
0.exit/back

1
```

- ④ Key in “1” (1. Boot_From_Others)

```
D:\workspace\sdk_release\WE2\V1.2\HimaxWE2_SDK_1.2_SVN184103\we2_ota_tool\ot...
=====OTA UPGRADE TOOL Version : 2.2.0=====
=====Other Tools Menu=====
Please choice following one item:
1.Boot from I2C (Application reboot to 2nd loader).....[BOOTROM]
2.Downloader FW via I2CBoot (Application).....[BOOTLOADER]
0.exit/back

1
Boot from I2C
=====File List=====
Please select file:
1. Boot_From_Others
2. Via_I2C_Boot
0. exit

1
=====File List=====
```

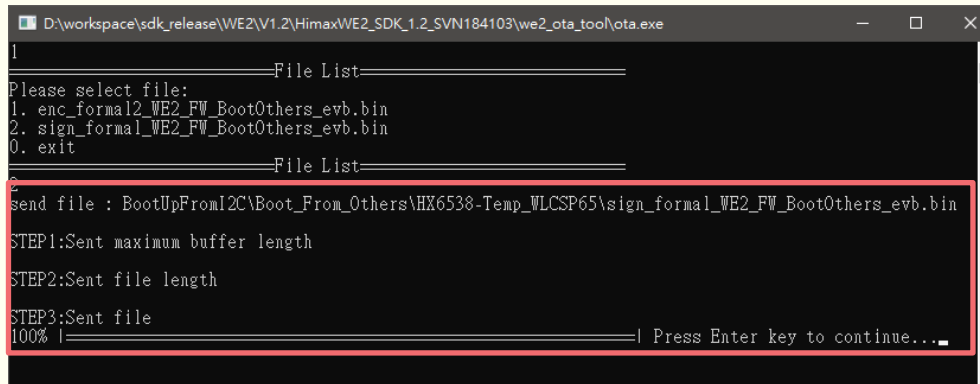
BootFromI2C Operations (3/4)

- ⑤ Key in “1” (1. HX6538-Temp_WLCSP65)
- ⑥ Key in “2” (2. sign_formal_WE2_FW_BootOthers_evb.bin)

```
D:\workspace\sdk_release\WE2\V1.2\HimaxWE2_SDK_1.2_SVN184103\we2_ota_tool\ot...
0. exit
=====File List=====
1
=====File List=====
Please select file:
1. HX6538-Temp_WLCSP65
0. exit
=====File List=====
1
=====File List=====
Please select file:
1. enc_formal2_WE2_FW_BootOthers_evb.bin
2. sign_formal_WE2_FW_BootOthers_evb.bin
0. exit
=====File List=====
2
```


BootFromI2C Operations (4/4)

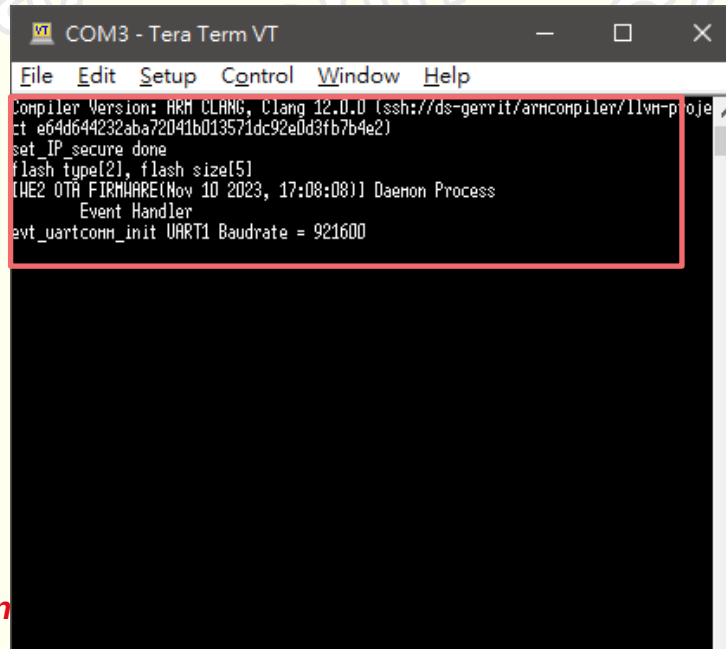
⑦ Wait the sent file complete



```
D:\workspace\sdk_release\WE2\V1.2\HimaxWE2_SDK_1.2_SVN184103\we2_ota_tool\ota.exe

1
=====File List=====
Please select file:
1. enc_formal2_WE2_FW_BootOthers_evb.bin
2. sign_formal_WE2_FW_BootOthers_evb.bin
0. exit
=====File List=====
2
send file : BootUpFromI2C\Boot_From_Others\HX6538-Temp_WLCSP65\sign_formal_WE2_FW_BootOthers_evb.bin
STEP1:Sent maximum buffer length
STEP2:Sent file length
STEP3:Sent file
100% |=====| Press Enter key to continue..._
```

⑧ The WE2 console(UART) will indicate it is ready for OTA upgrade.



```
COM3 - Tera Term VT
File Edit Setup Control Window Help
Compiler Version: ARM CLANG, Clang 12.0.0 (ssh://ds-gerrit/armcompiler/llvm-proje
ct e64d644232aba72041b013571dc92e0d3fb7b4e2)
set_IP_secure done
flash type[2], flash size[5]
[WE2 OTA FIRMWARE(Nov 10 2023, 17:08:08)] Daemon Process
Event Handler
evt_uartcomm_init UART1 Baudrate = 921600
```

OTA Upgrade Operations

OTA Upgrade Operations (1/4)

① Press any key to continue.

```
D:\workspace\sdk_release\WE2\V1.2\HimaxWE2_SDK_1.2_SVN184103\we2_ota_tool\ota.exe
=====File List=====
Please select file:
1. enc_formal2_WE2_FW_BootOthers_evb.bin
2. sign_formal_WE2_FW_BootOthers_evb.bin
0. exit
=====File List=====
2
send file : BootUpFromI2C\Boot_From_Others\HX6538-Temp_WLCSP65\sign_formal_WE2_FW_BootOthers_evb.bin
STEP1:Sent maximum buffer length
STEP2:Sent file length
STEP3:Sent file
100% |=====| Press Enter key to continue...-
```

② Key in “0” (0. exit/back)

```
D:\workspace\sdk_release\WE2\V1.2\HimaxWE2_SDK_1.2_SVN184103\we2_ota_tool\ota.exe
=====OTA UPGRADE TOOL Version : 2.2.0=====
=====Other Tools Menu=====
Please choice following one item:
1.Boot from I2C (Application reboot to 2nd loader).....[BOOTROM]
2.Downloader FW via I2CBoot (Application).....[BOOTLOADER]
0.exit/back
=====
0
```

OTA Upgrade Operations (2/4)

③ Key in “2” (2. Upgrade)

```
D:\workspace\sdk_release\WE2\V1.2\HimaxWE2_SDK_1.2_SVN184103\we2_ota_tool\ota.exe

=====OTA UPGRADE TOOL Version : 2.2.0 - Feb  1 2024=====
Transfer Protocol : I2C
=====OTA Main Menu=====
Please choice following one item:
1.Command
2.Upgrade
3.Others
0.exit

2_
```

④ Key in “1” (1. UPGRADE ALL IMAGE(Upgrade all image file))

```
D:\workspace\sdk_release\WE2\V1.2\HimaxWE2_SDK_1.2_SVN184103\we2_ota_tool\ota.exe

=====OTA UPGRADE TOOL Version : 2.2.0 - Feb  1 2024=====
Transfer Protocol : I2C
=====OTA Upgrade Menu=====
Please choice following one item:
1.UPGRADE ALL IMAGE(Upgrade all image file)
2.UPGRADE All APPLICATIONs
3.UPGRADE NS APPLICATIONs
0.exit/back

1
```

⑤ Key in “1” (1. output.img)

```
D:\workspace\sdk_release\WE2\V1.2\HimaxWE2_SDK_1.2_SVN184103\we2_ota...

1
Upgrade Image
=====File List=====
Please select file:
1. output.img
0. exit

1
```

OTA Upgrade Operations (3/4)

⑥ Wait the OTA upgrade...

```
D:\workspace\sdk_release\WE2\V1.2\HimaxWE2_SDK_1.2_SVN184103\we2_ota...
\count_length = 0x68b00
/count_length = 0x68c00
-count_length = 0x68d00
\count_length = 0x68e00
/count_length = 0x68f00
Getstatus
Device still Updating , waitting for device response!
read wrong feature(0x50) & type(0x3)
.Getstatus
status = 0x00120000

==>OTA_STATUS_OTA_RECEIVE_UPGRADE_DATA_DONE
status = 0x00120000

==>OTA_STATUS_OTA_RECEIVE_UPGRADE_DATA_DONE
.....
.....
```

⑦ OTA upgrade finished

```
D:\workspace\sdk_release\WE2\V1.2\HimaxWE2_SDK_1.2_SVN184103\we2_ota...
status = 0x00120000

==>OTA_STATUS_OTA_RECEIVE_UPGRADE_DATA_DONE
status = 0x00120000

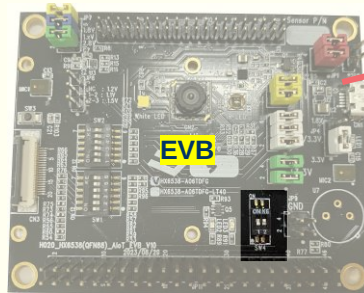
==>OTA_STATUS_OTA_RECEIVE_UPGRADE_DATA_DONE
...
Burn flash successss
Getstatus
status = 0x000c0000

==>OTA_STATUS_OTA_LOADER_IDLE
status = 0x000c0000

==>OTA_STATUS_OTA_LOADER_IDLE
ReBoot WE2
upgrade success
Press Enter key to continue...
```

OTA Upgrade Operations (4/4)

⑧ Set EVB to Normal mode



SWITCH SETTING	MODE
 1 2 [OFF, OFF]	EVB: Normal mode

⑨ Press Reset Button(SW3) to reboot WE2. The new Application should be run now.